

# Arghamitra Talukder

 ArghamitraT |  arghamitra.talukder@gmail.com |  +1.832.670.8231

## EDUCATION

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|-------------|-------------------------------------------------------------------|---------------------|
| 2023 - 2028 | Ph.D. in Computer Science, <b>Columbia University</b>             | New York, NY        |
|             | M.S. (2025) — <i>NSF CSGrad4US Fellow</i>                         | GPA: 3.98/4.0       |
| 2018 - 2021 | B.S. in Electrical Engineering at <b>Texas A&amp;M University</b> | College Station, TX |
|             | <i>Engineering Honors, Magna Cum Laude</i>                        | GPA: 3.78/4.0       |

## PROJECTS

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### Learning Interpretable Representations via Privileged-Basis Embeddings

Designed a training framework that enforces hierarchical information allocation across embedding dimensions, yielding a privileged basis with ordered feature importance and improved interpretability of latent representations.

### Predicting Exon Inclusion Levels via Contrastive Sequence Models

Developed a deep learning model to analyze gene regulation with a contrastive learning framework. This model identifies conserved functional elements by mapping evolutionarily related DNA sequences into a shared latent space, providing novel insights into gene expression.

### Bayesian hierarchical isoform quantification with hybrid sequencing

Developed JOLI, a novel hierarchical model that enhances RNA transcript quantification by jointly integrating short-read (SR) and long-read (LR) sequencing data. The model utilizes a multi-sample learning framework to improve estimation accuracy and reproducibility, particularly for low-abundance isoforms.

### Modeling cell plasticity with integrated single-cell data

Designed a statistical model to quantify cellular plasticity by integrating single-cell lineage and gene expression data. This approach utilizes a continuous annotation distribution methodology to statistically detect and score plastic cell states.

## PUBLICATIONS

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**Arghamitra Talukder**, Nicholas Keung, et al. (2026). “CLADES — Contrastive Learning Augmented Differential Splicing with Orthologous Positive Pairs”. In: *RECOMB (Accepted)*. [Online version](#).

**Arghamitra Talukder**, Shree Thavarekere, et al. (2025). “Multi-sample, multi-platform isoform quantification using empirical Bayes”. In: *RECOMB-Seq (Short-talk)*. [Online version](#).

**Arghamitra Talukder** and Itsik Pe’er (2024). “Computational Method to Detect Cancerous Plasticity”. In: *IICD Workshop on Single-Cell Data Integration & OT, CRA-WP Grad Cohort for IDEALS (Short-talk)*.

**Arghamitra Talukder**, Rujie Yin, et al. (2022). “Does inter-protein contact prediction benefit from multi-modal data and auxiliary tasks?” In: *MLSB, NeurIPS*. [Online version](#).

Ibrahim, Bassem, **Arghamitra Talukder**, and Roozbeh Jafari (2020). “Multi-source multi-frequency bio-impedance measurement method for localized pulse wave monitoring”. In: *EMBC, IEEE*. [Online version](#).

## WORK EXPERIENCE

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**Computational Engineering Intern, LLNL** June 2026 - Aug 2026

- *Lawrence Livermore National Laboratory (LLNL)*: Will develop machine learning methods for analysis of ALS-related datasets and evaluate model performance using statistical metrics.
- Will collaborate with interdisciplinary teams to support research planning, data analysis, and scientific discussions.

**Product & Test Engineer, Texas Instruments** June 2021 - Aug 2023

- Responsible for TI customized Ultra-High Voltage test technology with 70% cost efficiency.
- Created a product specific test program converting from legacy tester to the latest technology reducing 40% of the test time.
- Enabled traceability with the accuracy of 99.9% by device ID recovery obtained using OCR technology during internship.

## FELLOWSHIPS

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|                                     |                              |
|-------------------------------------|------------------------------|
| RECOMB Travel Fellowship            | May 2026                     |
| NIH T15 Training Grant              | Sep 2025 - Aug 2027          |
| CISE CSGrad4US Fellowship, CRA, NSF | Sep 2023 – Aug 2025 (Tenure) |
|                                     | Sep 2027 – Aug 2028 (Tenure) |

## TEACHING ASSISTANT

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|-------------------------------------------------|---------------------|
| Machine Learning for Functional Genomics        | Sep 2025 - Dec 2025 |
| Computational Genomics                          | Jan 2025 - May 2025 |
| Experimental Physics Lab II-Mechanics           | Jan 2019 - Dec 2019 |
| Foundation of Engineering in Python and LabVIEW | Sep 2018 - Dec 2018 |

## SKILLS

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|                       |                                                                                                             |
|-----------------------|-------------------------------------------------------------------------------------------------------------|
| Programming & ML      | Python, PyTorch, PyTorch Lightning, MATLAB, Perl, C, Bash                                                   |
| Genomics & Data       | Genomic sequence processing, RNA-seq (short- and long-read)                                                 |
| Systems & Cloud       | High-performance computing (SLURM, multi-GPU clusters), distributed training, Linux environments            |
| Engineering & Testing | Python/Perl/C-based semiconductor test program development, legacy tester migration and test time reduction |
| Tools & DevOps        | Git, Hydra/OmegaConf, WandB, Docker (basic)                                                                 |

## VOLUNTEERING

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|--------------------------------------------------------------|----------------|
| Symposium volunteer at IICD Next-Generation Cancer Research  | March, 2026    |
| Columbia pre-submission application reviewer                 | Nov 2025, 2024 |
| Conference volunteer at MLCB                                 | Sep 2025       |
| Panelist at the critical thinking in STEM career panel, NYGC | June 2024      |

## MENTORING

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|-------------------------------------|---------------------|
| Nicholas Keung, MS, Columbia CS     | Jan 2025 - Ongoing  |
| Abhinav Goel, Freshman, Columbia CS | Sep 2025 - Ongoing  |
| Tanya Mehta, MS, Columbia CS        | Sep 2025 - Dec 2025 |
| Shree Thavarekere, MS, Columbia CS  | Jan 2024 - Jan 2025 |
| Claire Dawson, MEng, Columbia CS    | Jan 2024 - May 2024 |